Project2 for Cloud Computing Fall 2026

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In this project, we develop a framework for weather monitoring. We use API from openweathermap.org. Users can enter how many days to forecast and zipcode. Weather data will be served through a Kafka channel. An evaluator will check whether certain alerts should be triggered, such as high or low temperature alerts, wind alerts, etc. Prometheus will be used for time-series data. Grafana will be used for any plots.

Golang will be the only language beyond the Kafka, Prometheus, and Grafana requirements. Your code needs to support multiple requests at once. These multiple requests could ask for different zip codes and different number of forecasted days.

AI use is permitted, however all prompts used need to be reported in the documentation. Also if used, AI name and version need to be reported.

The program needs to use go routines, concurrency, and channels. It should allow as many calls from different users at the same time and this feature should use the concurrency and channels of the same program. You can specify a specific way of calling your code multiple times at once and describe in your documentation.

The application needs to be hosted on Oracle cloud (https://www.oracle.com/cloud/free/) and this document needs cited there. All source code and files that document your test inputs/outputs should be submitted to Brightspace. Your submission will be compared to those of other students and your project score will be adjusted accordingly. Furthermore, the latency of your implementation and number of simultaneous requests may also affect your score.

Teams of one or two people are acceptable. The submission needs to be accompanied with a report that describes how to compile and run your code. Include input testcase files you may have used to ensure that your code is working properly. The report should discuss the following:

1. Software architecture used in the code and why these decisions were made.
2. Citations to previous work, such as papers or online work and discussions of how your work may be different. Please use an IEEE or ACM citation format either for a journal or conference paper, be consistent once you have picked your format.

3. Interactions will AI if AI is used.
4. Details on how you tested your implementation.

Including something surprising and better than other teams may unlock a bonus.

If you could not finish the project with all requirements such as introducing Kafka, Prometheus, or Grafana, it is still advisable to submit what you could accomplish.